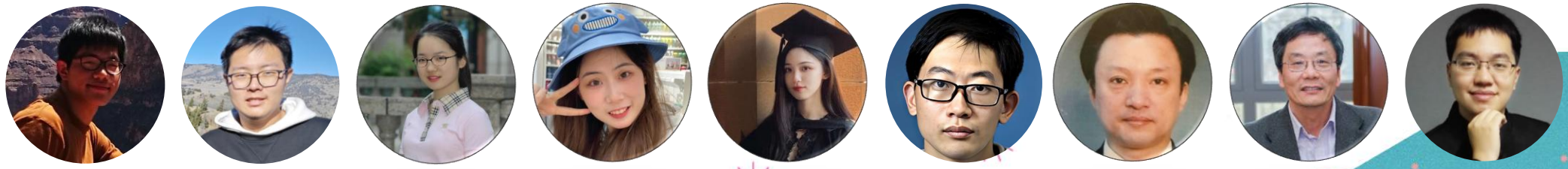


DTBIA: An Immersive Visual Analytics System for Brain-Inspired Research

Jun-Hsiang Yao , Mingzheng Li, Jiayi Liu, Yuxiao Li, Jielin Feng, Jun Han, Qibao Zheng, Jianfeng Feng, Siming Chen*



FUDAN UNIVERSITY

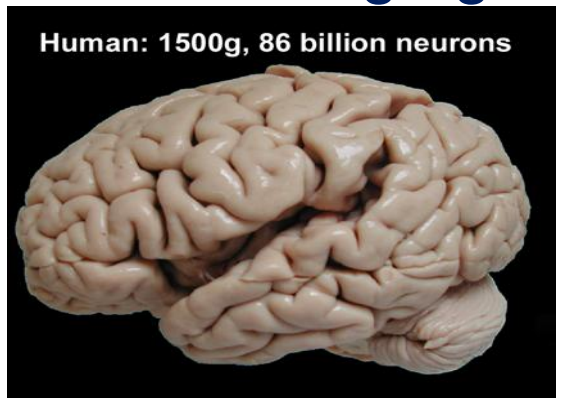
香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

THE OHIO STATE UNIVERSITY

Pacific Visualization 2025

What is the DTB?

Neuroimaging

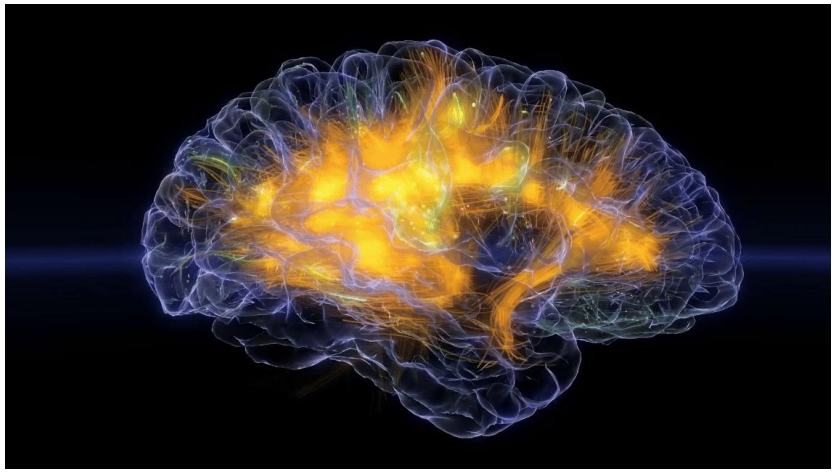


Distributed Computing

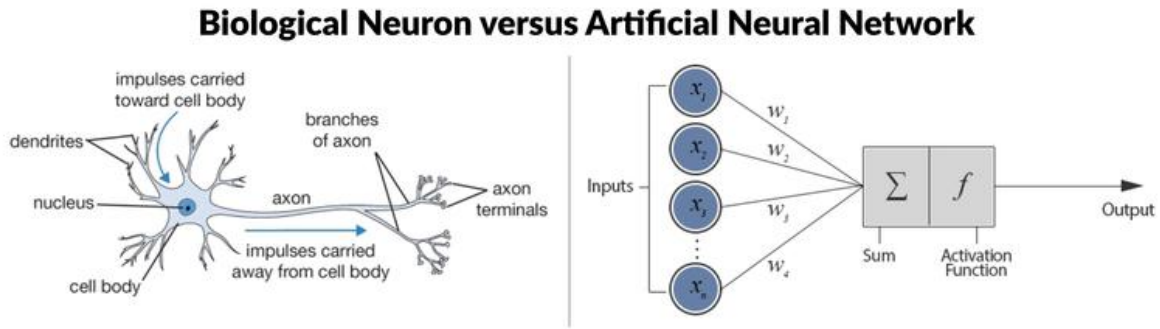


assimilation

Digital Twin Brain (DTB)



Mathematical Modeling



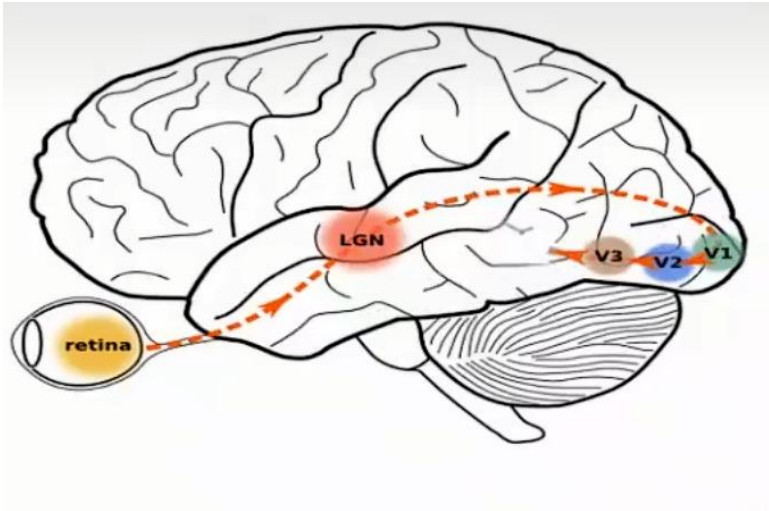
What DTB can do?

Neuroscience

- **Self Understanding**
  - Emotions
  - Perception
  - Cognition

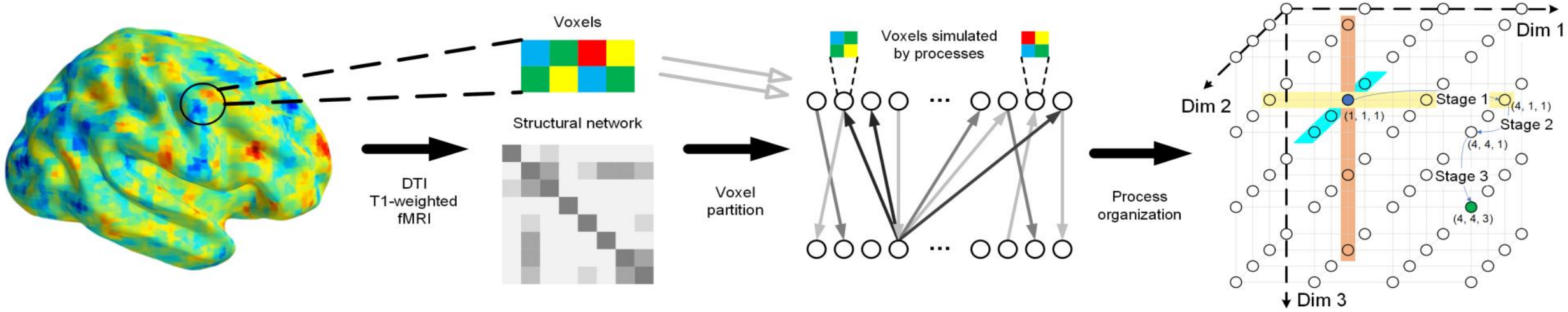
AI

- **Next Generation AI**
  - Automatic Car
  - Intelligent medicine
  - Adaptive robotics
  - …

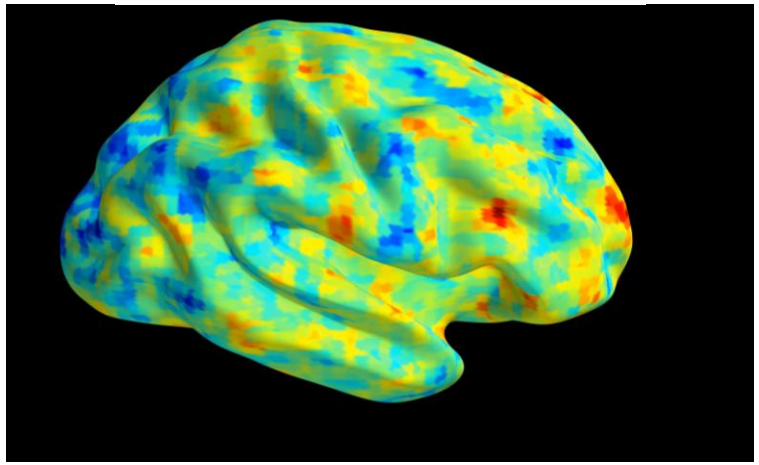




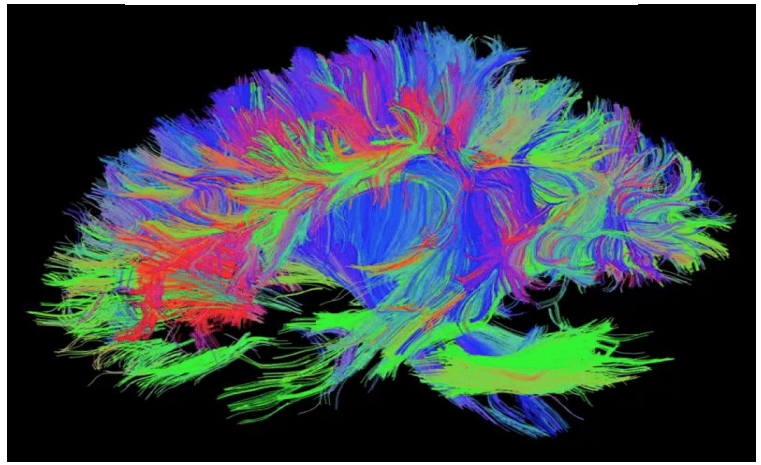
Bottom-Up Simulation Process



Functional Data

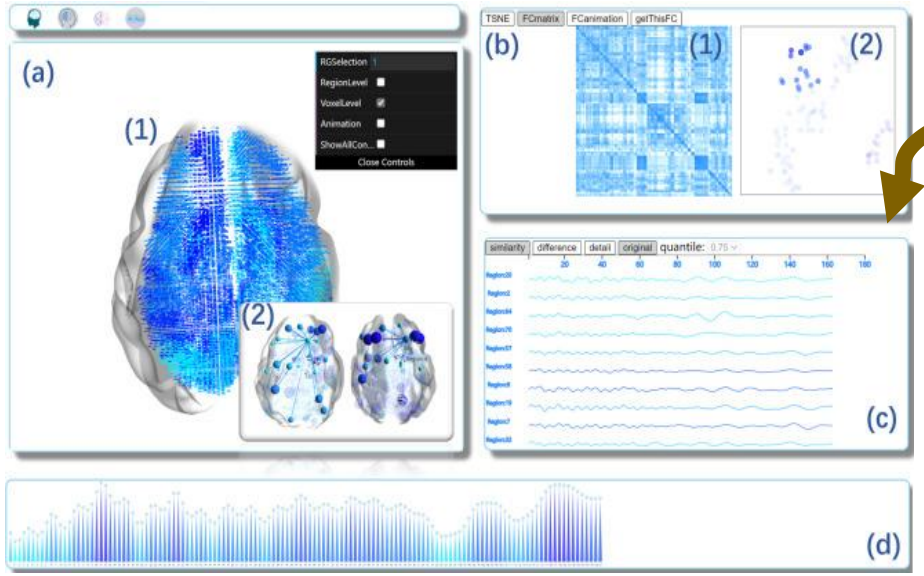


Structural Data

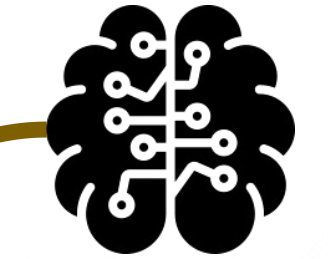


Motivation

Brain-Inspired Visualization



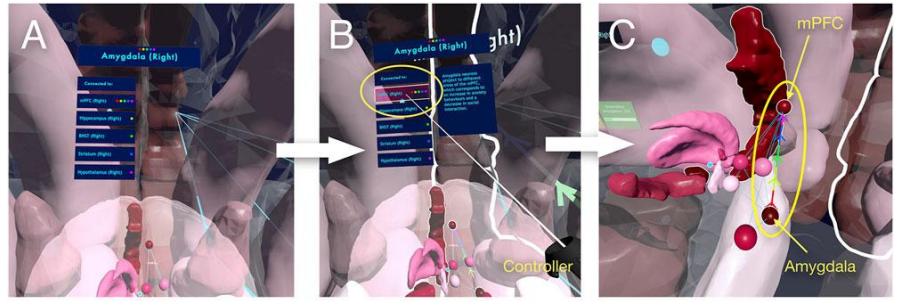
[**DTBVis**, Li et al., 2023]



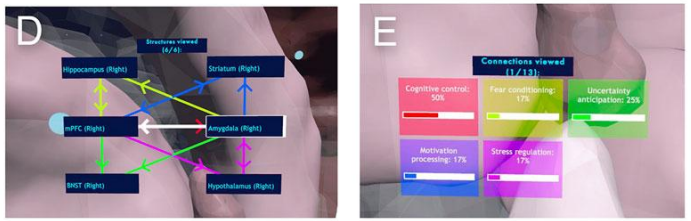


Immersive Environment

IA for brain on VR

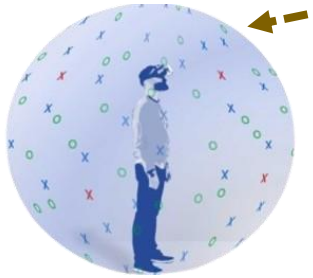


Workflow of selecting a connectivity

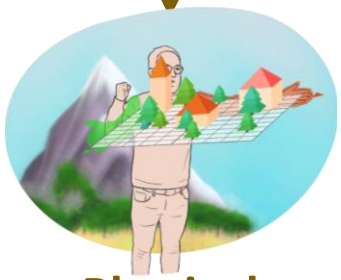


Connectivity diagram | Learning progress panel

[**SONIA**, Hellum et al., 2023]



Spatial Awareness



Physical Interaction

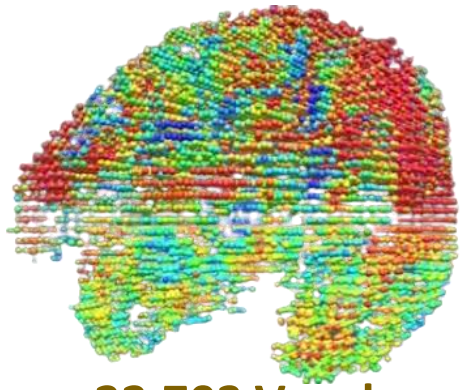


Enhanced Sense-making

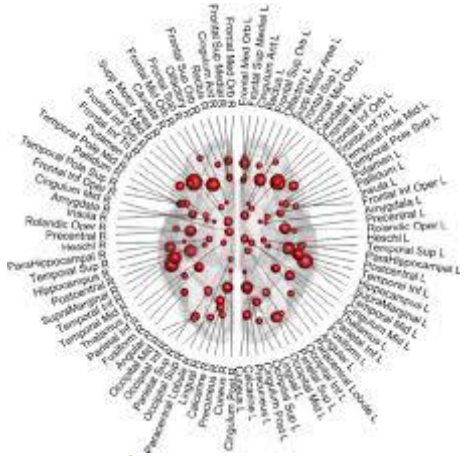

0
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Reset

Data Description

Brain Structure

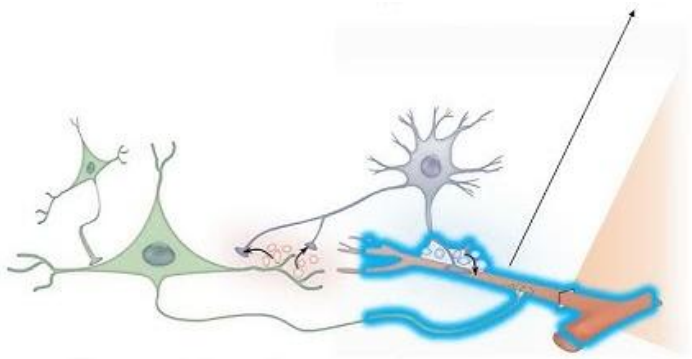


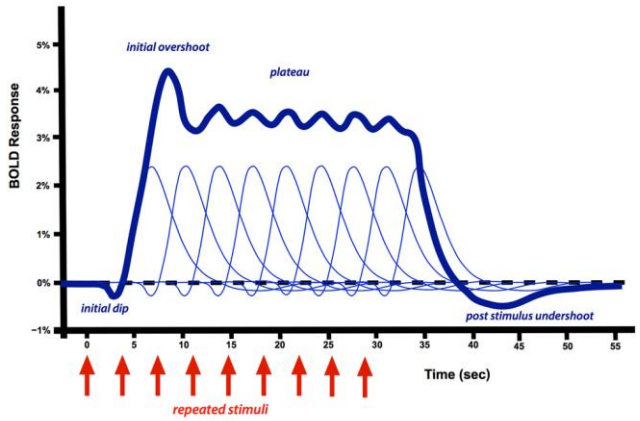
22,703 Voxels



92 brain regions
[AAL, Mazoyer et al., 2002]

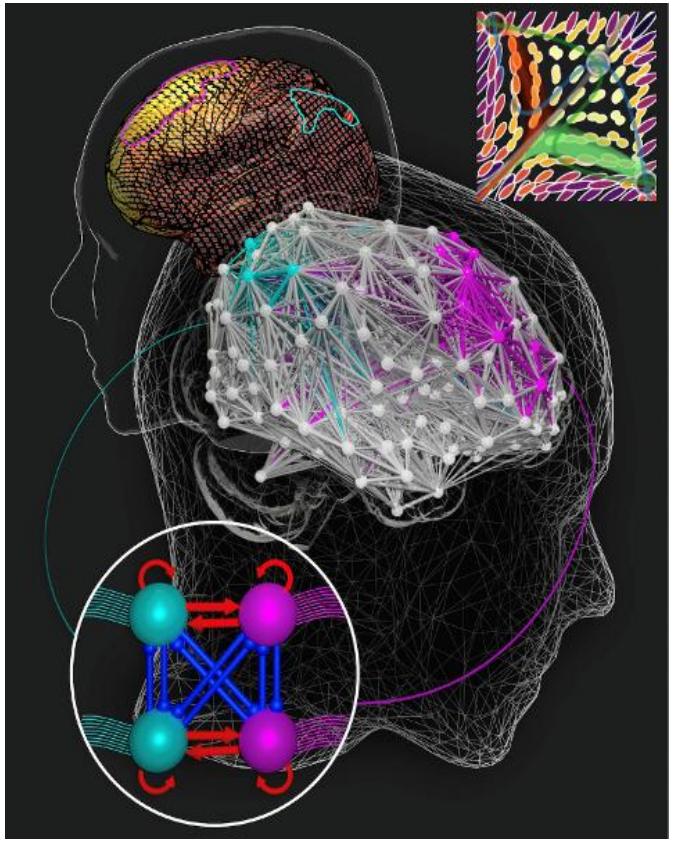
Functional Data





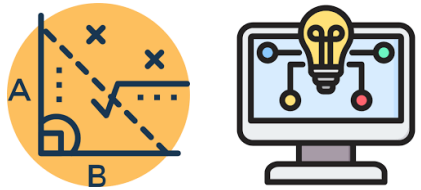
Sampled every 800 milliseconds over 166 time points for BOLD

Structural Data



22,703 x 22,703 fully connected matrix with DTI scans

Collaborative Expert Needs



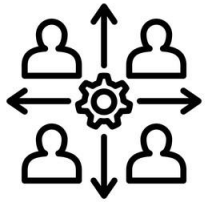


8 Math & CS Experts

1. Validate DTB vs biological brain
2. Identify model failure patterns
3. Optimize simulated BOLD dynamics

2 Neuroscientists

1. Collect Region of Interests (RoIs)
2. Trace structure-based RoIs chains
3. Highlight task-related activation bursts



Shared Cross-Team Needs

1. Top-down analysis workflow
2. Unified structure function views

Task Analysis

T1. Signal Similarity

- Compare DTB-generated BOLD traces with real fMRI to quantify model fidelity.

T2. Spatiotemporal Function

- Trace ROI activity across time and scale to reveal functional patterns.

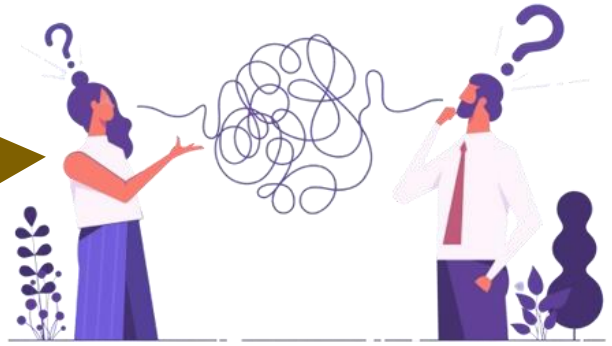
T3. Structural Topology

- Map and analyze voxel-to-region connectivity to expose critical pathways.

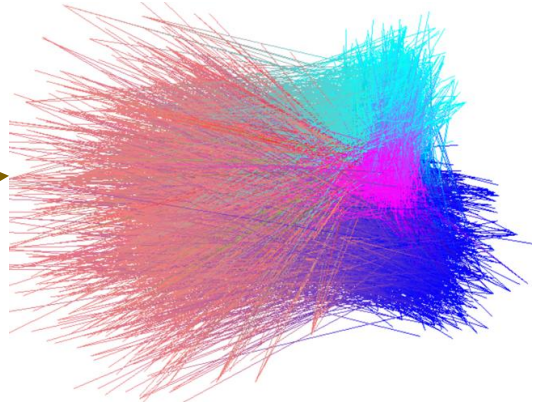
9

Challenges

1. **Cross Domain Communication** → **Inadequate Spatial Awareness**



2. **Insight extraction** → **Overwhelming Visual Complexity**

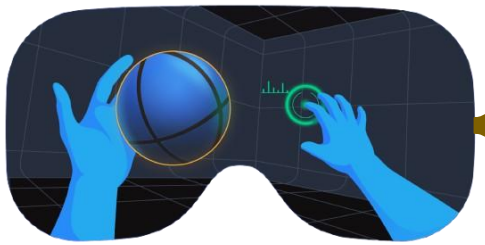


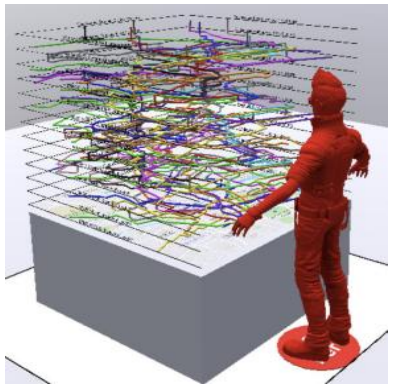
3. **Pattern discovery** → **Inefficient Multi Scale Exploration**



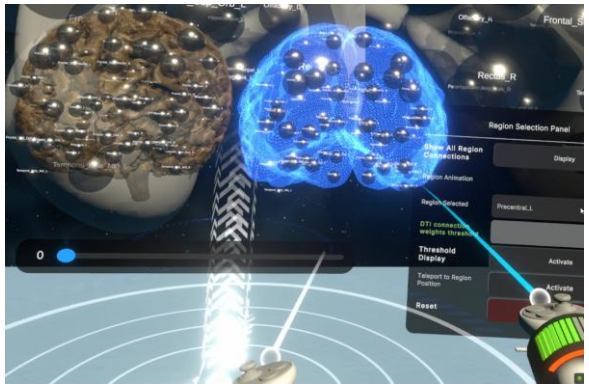
Immersive Degree

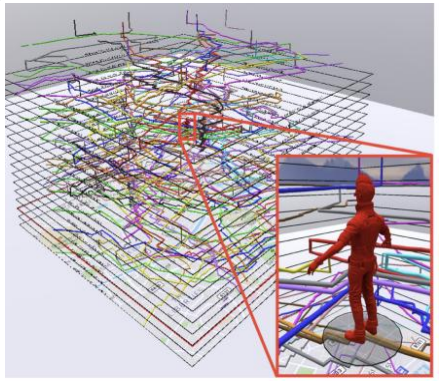
1 Cross Domain Communication → Inadequate Spatial Awareness



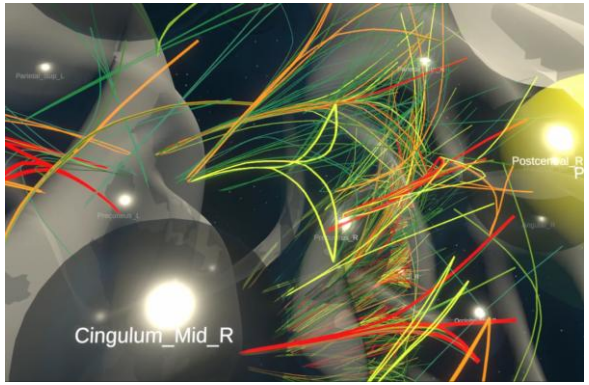


Exocentric + Walking → Real Scale Brain



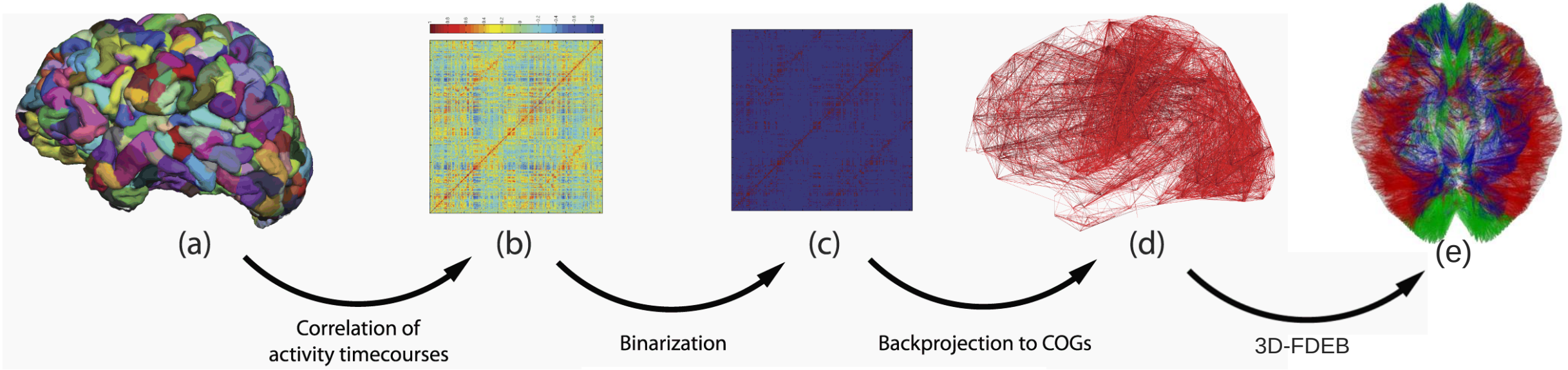


Egocentric + Flying → Large Scale Brain

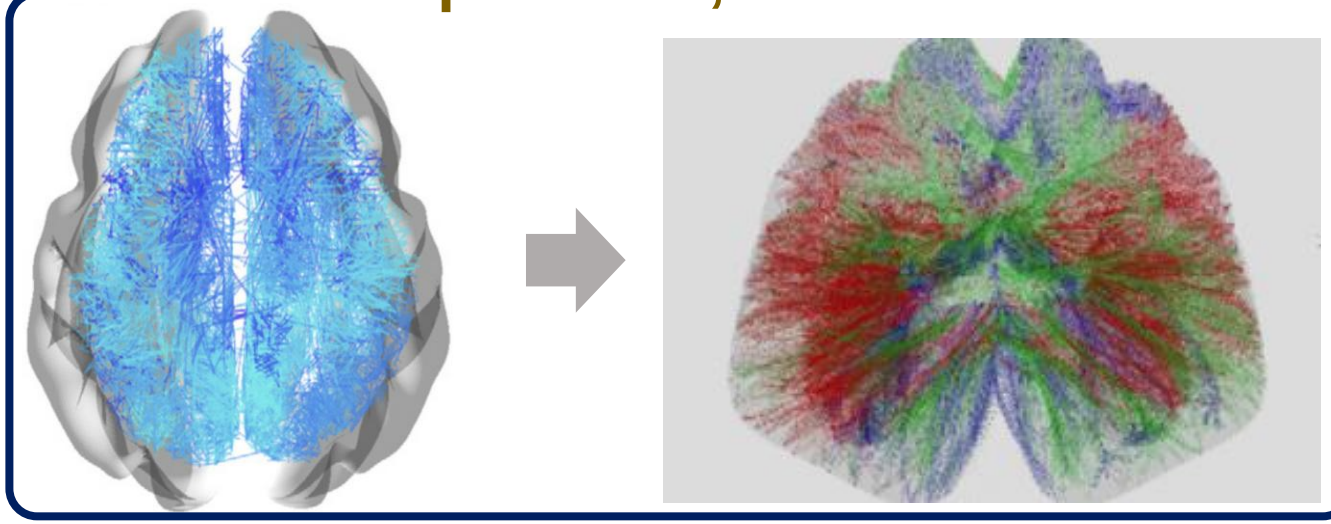


3D-FDEB for DTI

❷ Insight extraction ➡ Overwhelming Visual Complexity

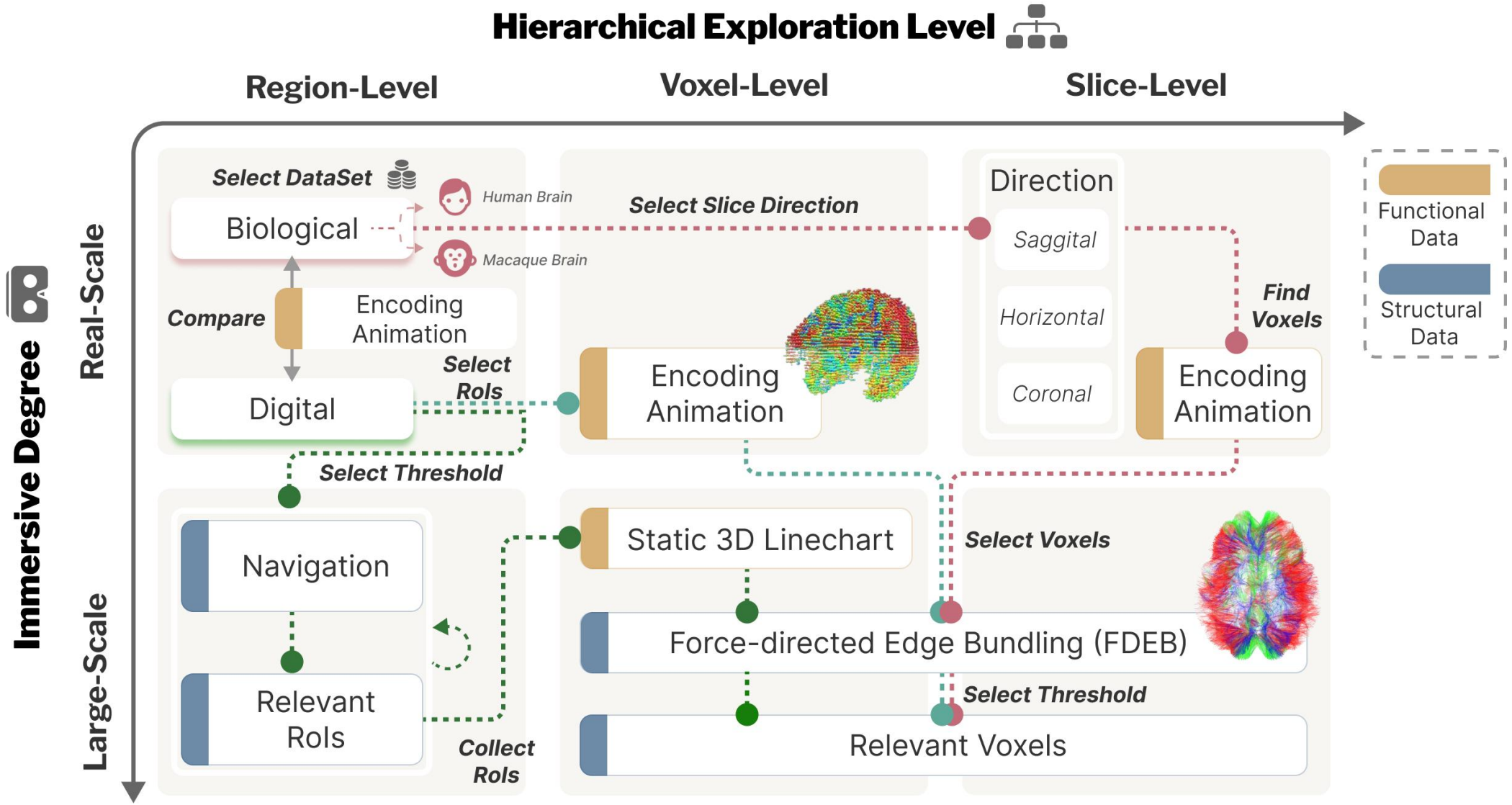


Top 10% 38,036 links

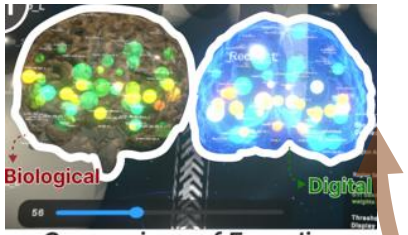


Top-down Workflow

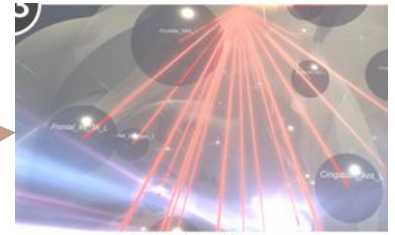
3 Pattern discovery → Inefficient Data-Scale Exploration



Region Level Exploration



Comparison of Encoding Animation


Navigation

Hierarchical Exploration Level

Region-Level | Voxel-Level | Slice-Level

Immersive Degree

Real-Scale

Large-Scale

Select DataSet

Human Brain

Macaque Brain

Biological

Compare

Encoding Animation

Digital

Select Rols

Select Threshold

Navigation

Relevant Rols

Collect Rols

Select Slice Direction

Direction

Saggital

Horizontal

Coronal

Find Voxels

Encoding Animation

Encoding Animation

Static 3D Linechart

Select Voxels

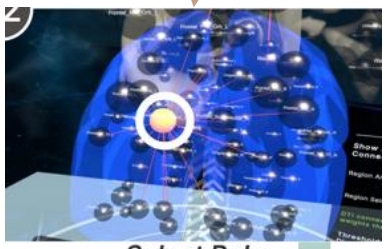
Force-directed Edge Bundling (FDEB)

Select Threshold

Relevant Voxels

Functional Data

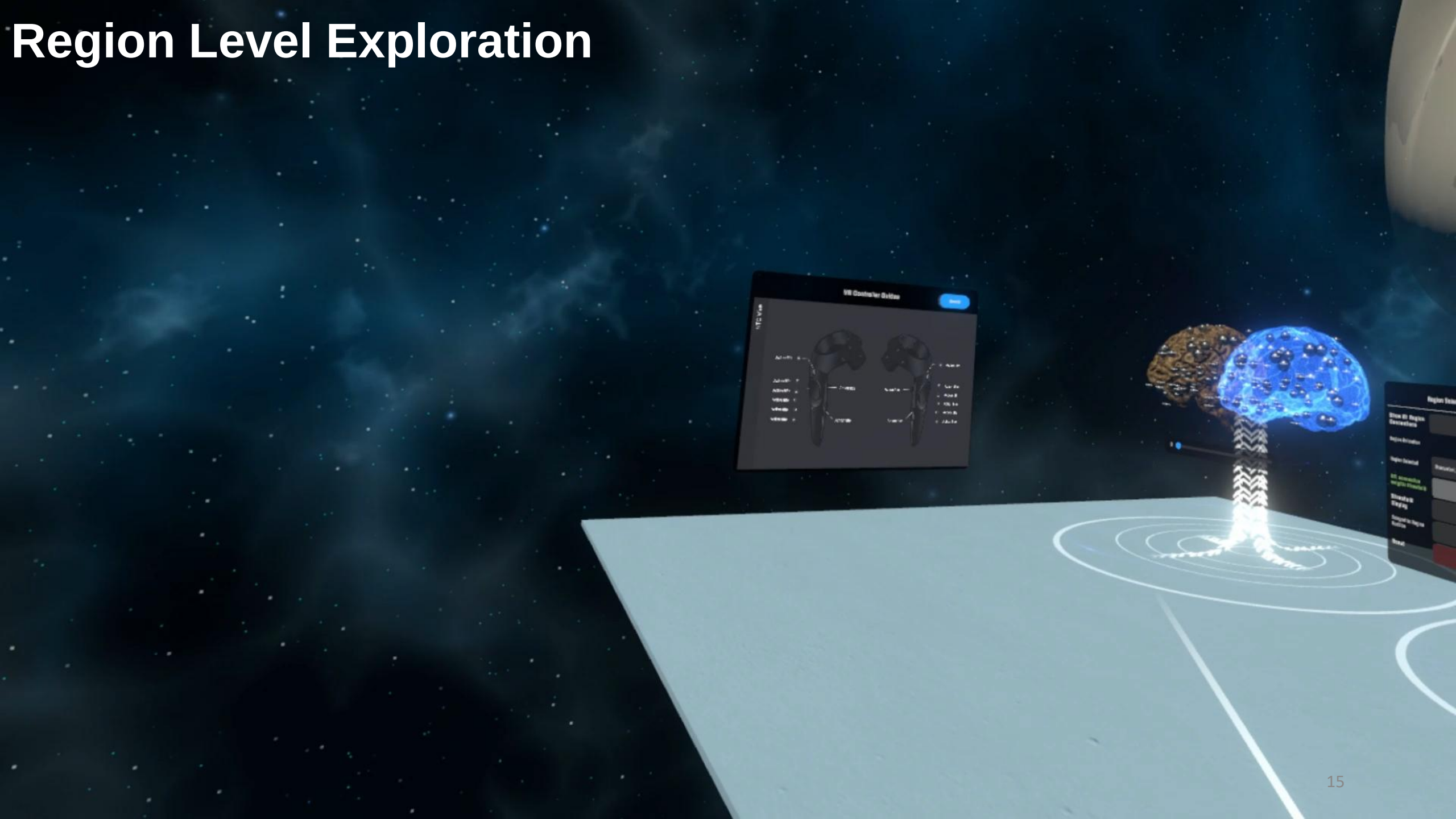
Structural Data


Select Rols

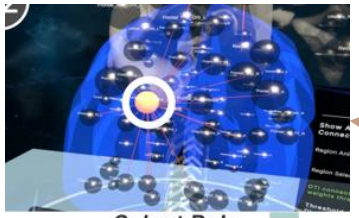

Relevant Rols


Collected Rols

Region Level Exploration

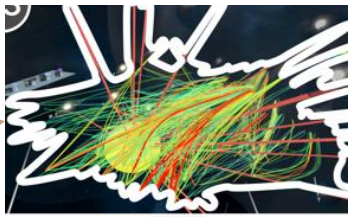


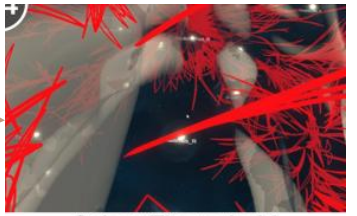
Voxel Level Exploration

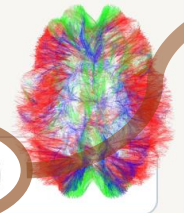

Select Rols

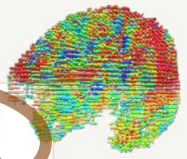

Voxel-Level Encoding Animation


Static 3D Linechart


FDEB


Select Thereshold





Hierarchical Exploration Level

Region-Level | Voxel-Level | Slice-Level

Real-Scale

Large-Scale

Immersive Degree

Select DataSet

Human Brain

Macaque Brain

Biological

Digital

Compare

Encoding Animation

Select Rols

Select Threshold

Navigation

Relevant Rols

Collect Rols

Select Slice Direction

Encoding Animation

Static 3D Linechart

Force-directed Edge Bundling (FDEB)

Relevant Voxels

Select Voxels

Select Threshold

Direction

Saggital

Horizontal

Coronal

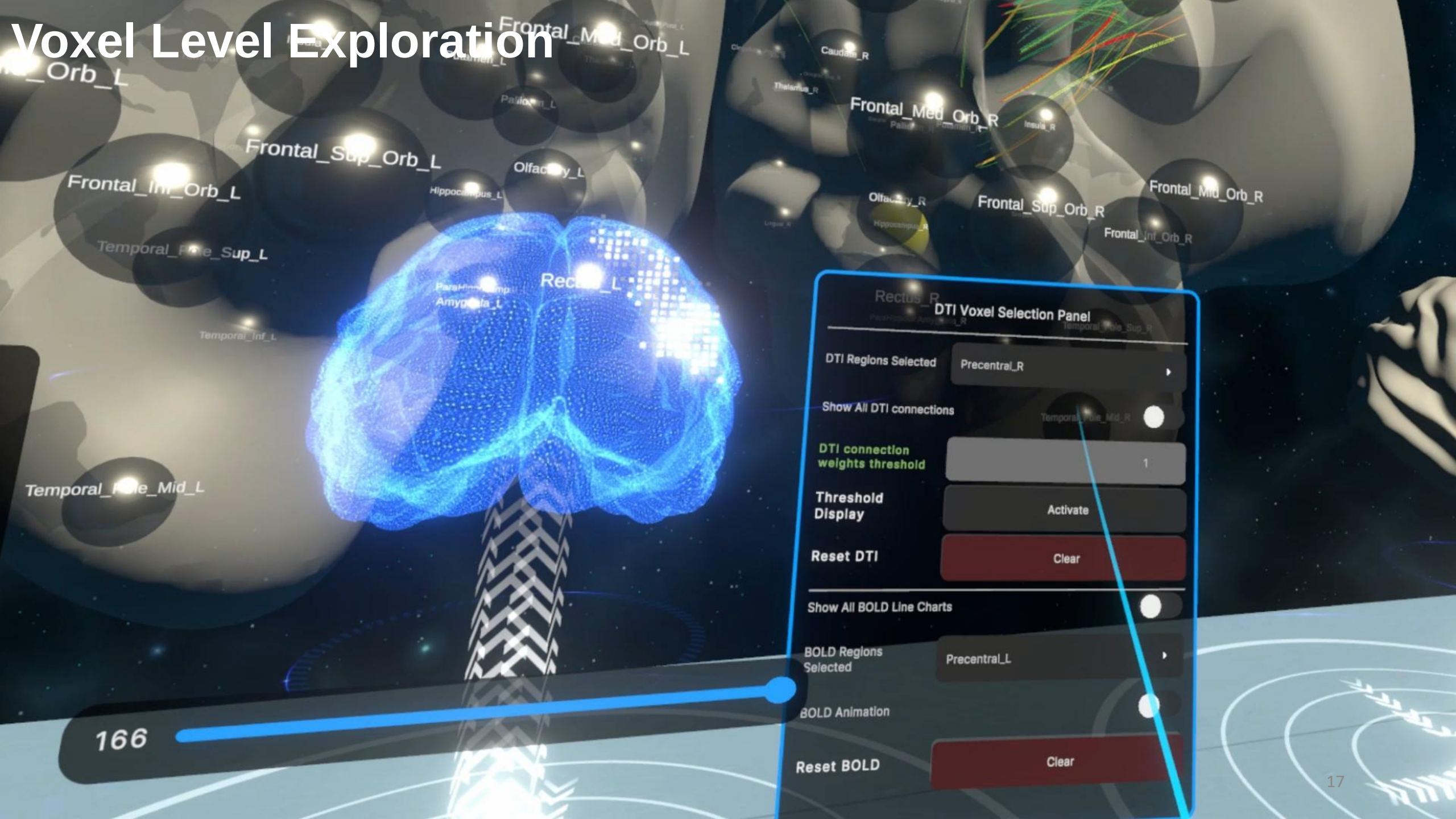
Find Voxels

Encoding Animation

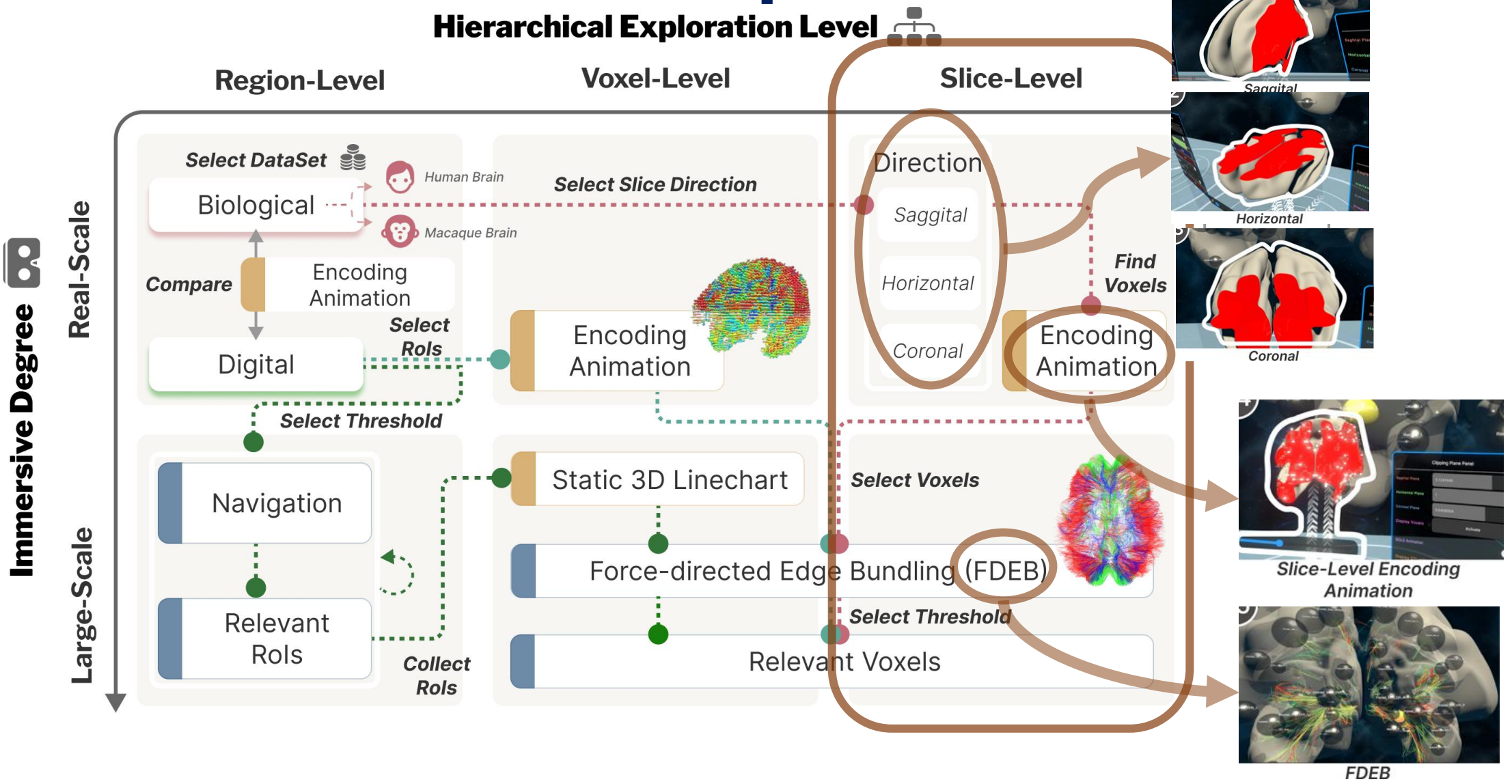
Functional Data

Structural Data

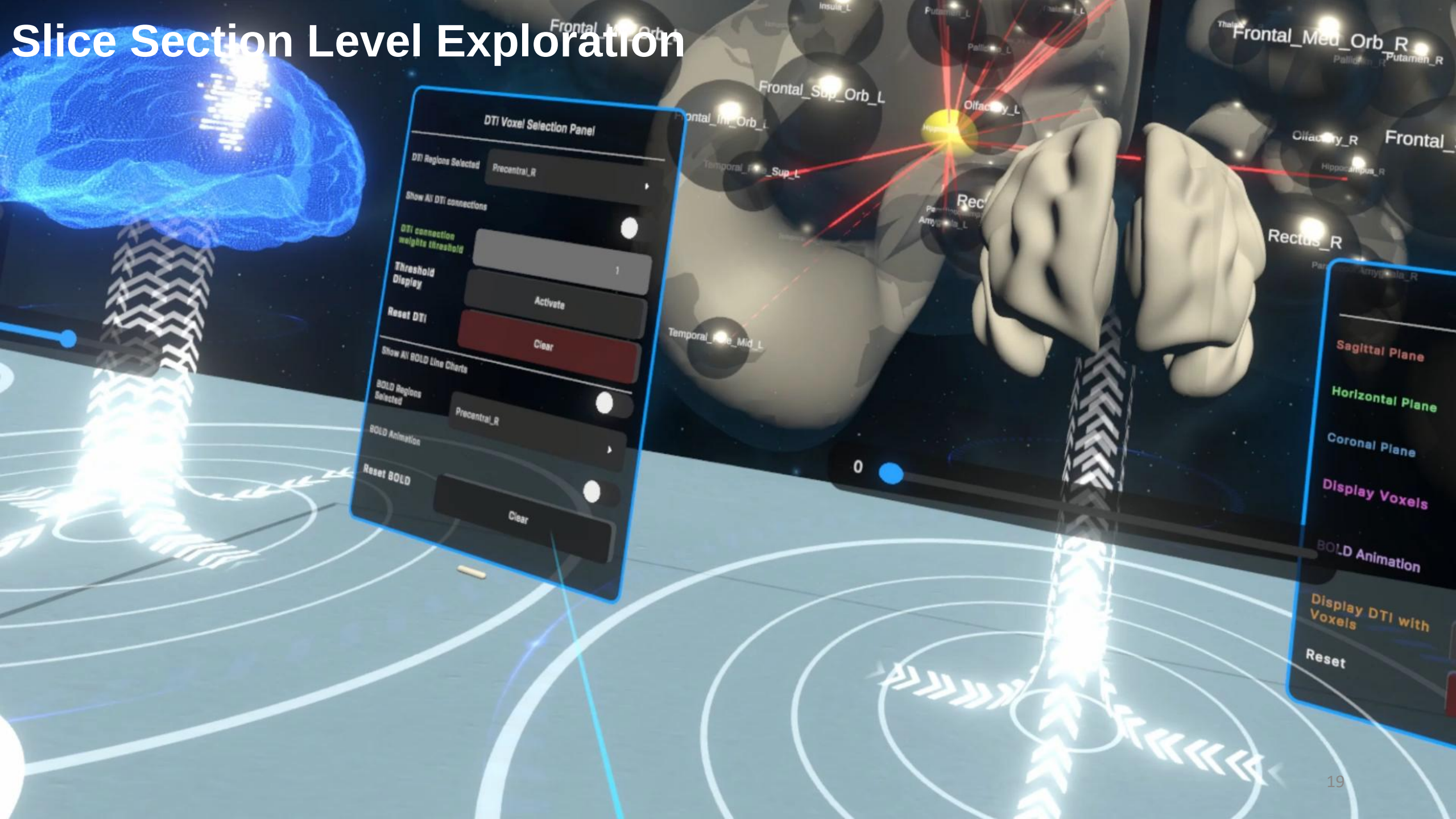
Voxel Level Exploration



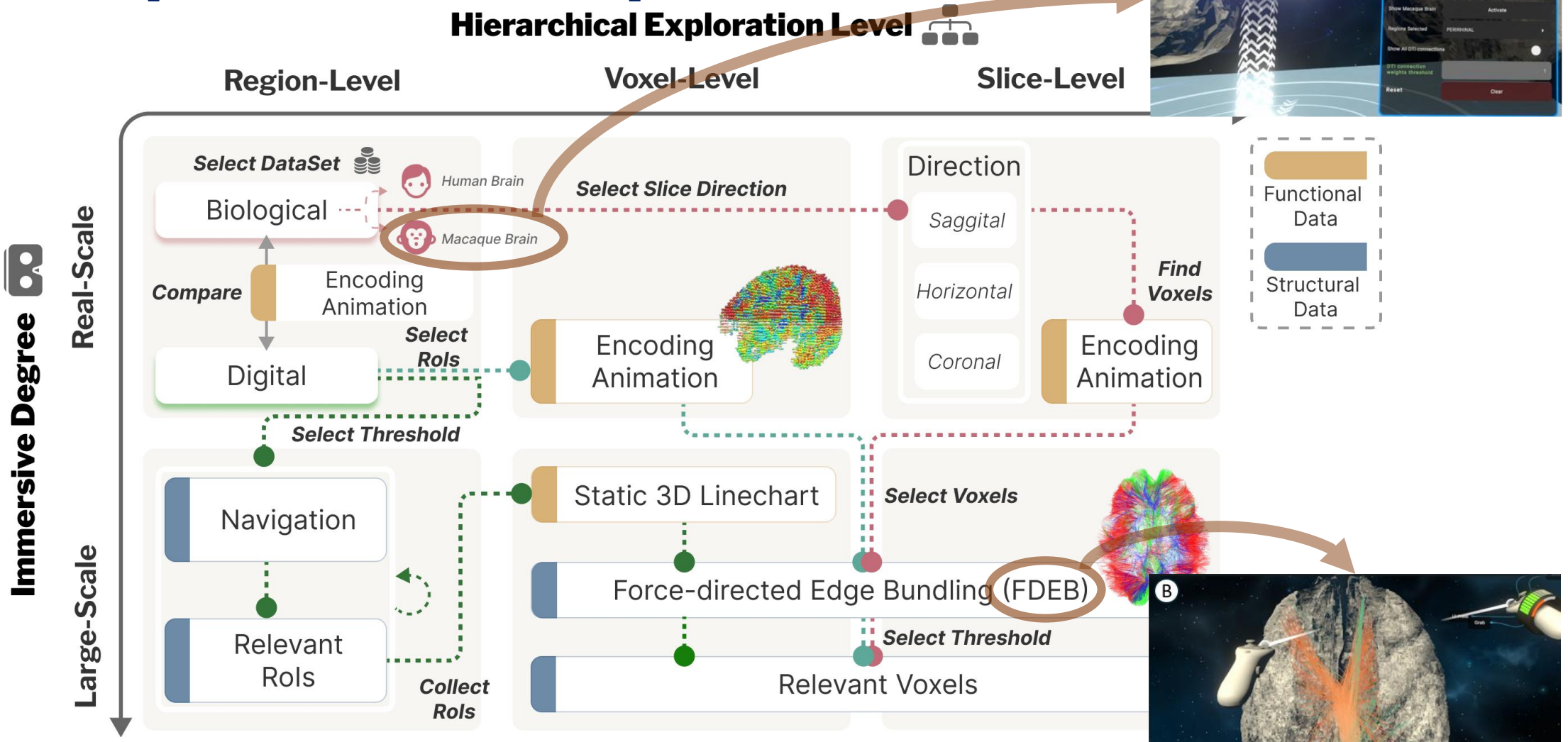
Slice Section Level Exploration



Slice Section Level Exploration



Macaque Brain Exploration



Validation: 2 Case Studies

Case 1: no biological expertise
Case 2: with biological expertise
2 experts in-HMD; 10 experts watching their VR feed on a shared screen.

Goals:
1. Utility and Effectiveness of the DTBIA system
2. Assess the user experience in immersive environments
3. Implications for real-world usages



Step 1
Introduction

Step 2
Training

Step 3
Free Exploration + Think-Aloud

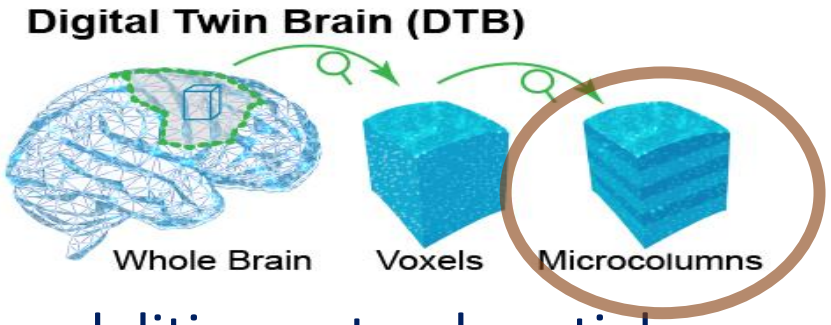
Step 4
Interview

Study Results

1. Experts could conduct **effective DTB analysis** with DTBIA
   - Drill-down speeds ROI detection.
   - Reveals DMN patterns and confirms hypotheses.
   - High **learnability**, **utility**, and **satisfaction**.

2. Immersive 3-D greatly enhances **spatial insight** & **collaboration**
   - Egocentric navigation improved **perception** of brain topology and ROI positions.
   - Shared VR context **streamlining** dialogue between AI experts and neuroscientists.
   - Immersive 3D visualizations deepen **understanding** and user **engagement**.

Future Work



1. Richer Multimodal Data

- Integrate EEG/MEG and other modalities; extend spatial resolution to cortical micro-columns; include task-based states (e.g., auditory, visual).



2. Real-Time Multi-User Collaboration

- Add multiplayer VR support.

3. Comfort-Oriented Locomotion

- Refine navigation modes (e.g., vignette, snap-turn, adaptive speed) to minimize cybersickness during extended sessions.



Takeaway

1. Co-designed with 12 experts, **bridging** brain-inspired computing & neuroscience to **advance DTB research**.

2. Developed an immersive visual analytics **platform** for DTB spatiotemporal data that **streamlines cross-domain communication**.

3. **Case studies and expert feedbacks** confirm DTBIA's real-world value.

DTBIA: An Immersive Visual Analytics System for Brain-Inspired Research


Jun-Hsiang Yao


Mingzheng Li


Jiayi Liu


Yuxiao Li


Jielin Feng


Jun Han


Qibao Zheng


Jianfeng Feng


Siming Chen

Thank you!

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Siming Chen simingchen@fudan.edu.cn

FUDAN UNIVERSITY

ISTBI 复旦大学类脑智能科学与技术研究院 Institute of Science and Technology for Brain-Inspired Intelligence

香港科技大學 THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

THE OHIO STATE UNIVERSITY